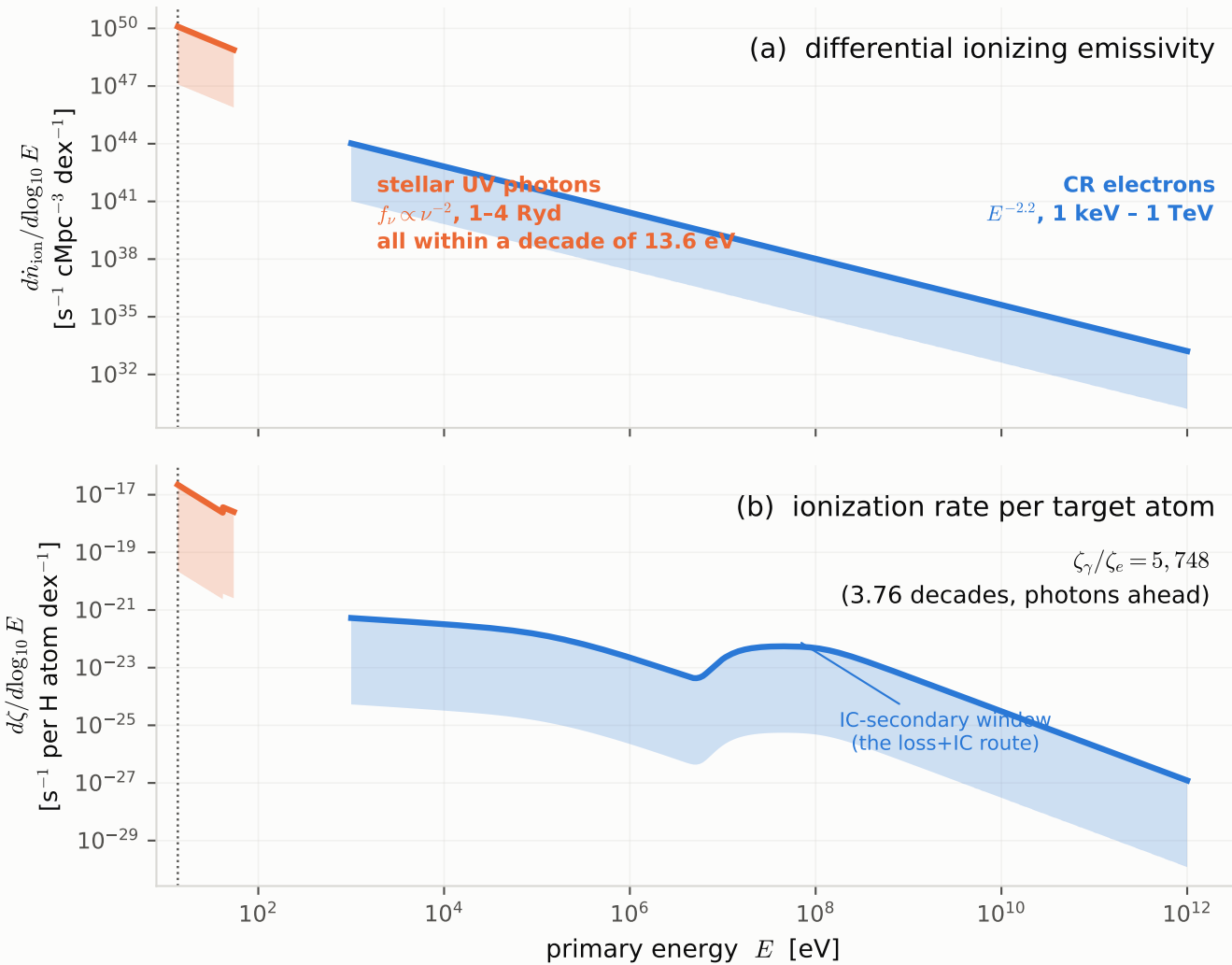


Ionizing emissivity and ionization rate at $z = 20$ canonical parameters



IGM at $z=20$, $x_e=10^{-4}$, pure H.
 Electron yields: the loss+IC route — igm_losses.py: 7 loss mechanisms, RBED event counting, BED secondary cascade, Klein-Nishina IC, + IC-seconda
 ρ_{UV} : Donnan+24 JWST PRIMER. $\xi_{\text{ion}}=10^{25.28}$: Llerena+25. $\rho_{\text{UV}} \rightarrow \rho_{\text{SFR}}$: Madau & Dickinson 14 (Salpeter).
 $\text{SN}/\text{M}_{\odot}$ derived from the IMF, not adopted.
 Stellar SED: population of U37126 ($z=10.255$, Marques-Chaves+26, BPASS v2.2.1, $Z=0.003$, 6.8 Myr); LyC shape parametrised, α bracketed 1-3.
 NOT ANCHORED: ρ_{UV} and ξ_{ion} are measured at $z \simeq 10$, not $z=20$. The shaded band spans $\rho_{\text{SFR}} \in [0.001, 1] \times$ its $z=10$ value — an ignorance interval, not a